

Deepak Mallampati

APPLIED AI ENGINEER - GENAI & FINE-TUNING

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US-based, F-1 OPT - open to EU relocation; authorization to be discussed

PROFESSIONAL SUMMARY

Applied AI engineer shipping production GenAI end-to-end - LLM fine-tuning, RAG, and agents - with evaluation rigor. Fine-tuned Qwen3-27B via 4-bit QLoRA on an NVIDIA H100, delivered banking/healthcare RAG at 92% precision (~40% hallucination), and built a conductor-subagent agent framework (~42% tokens). Evidence-based model selection across GPT-4o/Claude/Gemini (+27% task success). Strong fit for deploying high-impact GenAI for enterprise clients and contributing to open-source inference/fine-tuning.

CORE COMPETENCIES & SKILLS

Applied AI & Forward Deployed • LLM Fine-tuning (LoRA, QLoRA, PEFT) • RAG & Retrieval • Production GenAI • Agentic Systems • Model Evaluation • Hugging Face Transformers • Enterprise Deployment

Generative AI & LLMs: GPT-4o, GPT-5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | RAG Architecture | Prompt Engineering | Multi-Agent Systems | Fine-tuning (LoRA, QLoRA, PEFT) | Content Safety | Hugging Face Transformers

Machine Learning & NLP: PyTorch, TensorFlow, Keras | NER, Text Classification, Summarization | Model Optimization | A/B Testing | Drift Detection | RAGAS Evaluation | BERTScore

Vector Search & Retrieval: FAISS, Pinecone, Weaviate | Semantic & Hybrid Search | Cross-Encoder Re-ranking | Document Chunking | Semantic Caching

MLOps & Cloud: AWS (EC2, S3, Lambda, Bedrock, SageMaker), Azure (OpenAI, AI Services), GCP Vertex AI | Docker, Kubernetes | Terraform | CI/CD | MLflow, Weights & Biases | Model Monitoring & Observability

Data Engineering: PostgreSQL, MongoDB, Redis | Spark, Databricks | Kafka | Pandas, NumPy | Data Lineage & Governance

Security & Compliance: HIPAA, SOC 2, PCI DSS | OAuth 2.0, JWT, RBAC | PII Masking/Redaction | Prompt Injection Mitigation | Audit Logging

Languages & Tools: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | Git, Jira, Confluence | Agile/Scrum

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present

AI Engineer

USA

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing objectives into context-scoped subagents - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift relevance and materially reduce hallucination.
- Established a structured **LLM evaluation & monitoring framework** (RAGAS, BERTScore, custom metrics) with latency/accuracy dashboards to catch regressions before release.
- Optimized inference cost and latency through prompt compression, **semantic caching, and model routing**, sustaining SLA targets while reducing per-query overhead.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retries, reducing failed production runs by **60%**.

Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern

USA

- Developed **AI agents and backend services** for Vanguard's internal AI platform, integrating with production data pipelines and observability across **25+ AI agents** and workflows.
- Optimized prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** and informing platform model-selection.
- Built **12 APIs and microservices**, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by **40%**.
- Engineered safeguards and content controls, reducing unsafe outputs by **42%**; partnered across 20+ teams supporting **100,000+ requests daily**.

Kent State University

Oct 2023 - Jan 2024

ML & Gen AI Research Assistant

USA

- Built a **privacy-preserving ML pipeline** (k-anonymity, l-diversity, Laplace differential privacy), cutting re-identification risk from 3.38% to 0.32% while retaining **99% of baseline predictive performance**.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 with a curated instruction dataset for controllable long-form generation; authored an IEEE-format conference paper.

Emerson

Jun 2022 - Apr 2023

ML Trainee

India

- Trained **regression and classification models** for equipment-failure prediction, anomaly detection, and capacity forecasting on operational sensor data.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification, achieving **89% F1** on custom NER tasks.

PROJECTS

MangaLens

[Live on Chrome Web Store](#)

Shipped Chrome extension delivering real-time AI translation of manga and webtoons across **29+ sites**, turning a multi-step manual chore into a single inline action.

career-ops

Autonomous, AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision.

Privacy-Preserving Clinical ML Pipeline

Framework combining k-anonymity, l-diversity, and differential privacy to analyze sensitive patient data without exposing identities - quantifying the privacy/accuracy trade-off.

Story Director LLM

Fine-tuned model generating structured short-form video narratives from a single prompt, paired with an automated rendering pipeline.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 | GPA 3.70/4.0